

An Analysis of Algorithmic bias

Algorithmic bias

Algorithmic bias -- Part 1

Gender discrimination In 2016, the professional networking site LinkedIn was discovered to recommend male variations of women's names in response to search queries. The site did not make similar recommendations in searches for men's names. For example, "Andrea" would bring up a prompt asking if users meant "Andrew", but queries for "Andrew" did not ask if users meant to find "Andrea". The company said this was the result of an analysis of users' interactions with the site. In 2012, the department store franchise Target was cited for gathering data points to infer when female customers were pregnant, even if they had not announced it, and then sharing that information with marketing partners. Because the data had been predicted, rather than directly observed or reported, the company had no legal obligation to protect the privacy of those customers. Web search algorithms have also been accused of bias. Google's results may prioritize pornographic content in search terms related to sexuality, for example, "lesbian". This bias extends to the search engine showing popular but sexualized content in neutral searches. For example, "Top 25 Sexiest Women Athletes" articles displayed as first-page results in searches for "women athletes". In 2017, Google adjusted these results along with others that surfaced hate groups, racist views, child abuse and pornography, and other upsetting and offensive content. Other examples include the display of higher-paying jobs to male applicants on job search websites. Researchers have also identified that machine translation exhibits a strong tendency towards male defaults. In particular, this is observed in fields linked to unbalanced gender distribution, including STEM occupations. In fact, current machine translation systems fail to reproduce the real world distribution of female workers. In 2015, Amazon.com turned off an AI system it developed to screen job applications when they realized it was biased against women. The recruitment tool excluded applicants who attended all-women's colleges and resumes that included the word "women's". A similar problem emerged with music streaming services—In 2019, it was discovered that the recommender system algorithm used by Spotify was biased against female artists. Spotify's song recommendations suggested more male artists over female artists.

Algorithmic bias -- Part 2

Complexity Algorithmic processes are complex, often exceeding the understanding of the people who use them. Large-scale operations may not be understood even by those involved in creating them. The methods and processes of contemporary programs are often obscured by the inability to know every permutation of a code's input or output. Social scientist Bruno

Latour has identified this process as blackboxing, a process in which "scientific and technical work is made invisible by its own success. When a machine runs efficiently, when a matter of fact is settled, one need focus only on its inputs and outputs and not on its internal complexity. Thus, paradoxically, the more science and technology succeed, the more opaque and obscure they become." Others have critiqued the black box metaphor, suggesting that current algorithms are not one black box, but a network of interconnected ones. An example of this complexity can be found in the range of inputs into customizing feedback. The social media site Facebook factored in at least 100,000 data points to determine the layout of a user's social media feed in 2013. Furthermore, large teams of programmers may operate in relative isolation from one another, and be unaware of the cumulative effects of small decisions within connected, elaborate algorithms. Not all code is original, and may be borrowed from other libraries, creating a complicated set of relationships between data processing and data input systems. Additional complexity occurs through machine learning and the personalization of algorithms based on user interactions such as clicks, time spent on site, and other metrics. These personal adjustments can confuse general attempts to understand algorithms. One unidentified streaming radio service reported that it used five unique music-selection algorithms it selected for its users, based on their behavior. This creates different experiences of the same streaming services between different users, making it harder to understand what these algorithms do. Companies also run frequent A/B tests to fine-tune algorithms based on user response. For example, the search engine Bing can run up to ten million subtle variations of its service per day, creating different experiences of the service between each use and/or user.

Algorithmic bias -- Part 3

The earliest computer programs were designed to mimic human reasoning and deductions, and were deemed to be functioning when they successfully and consistently reproduced that human logic. In his 1976 book *Computer Power and Human Reason*, artificial intelligence pioneer Joseph Weizenbaum suggested that bias could arise both from the data used in a program, but also from the way a program is coded. Weizenbaum wrote that programs are a sequence of rules created by humans for a computer to follow. By following those rules consistently, such programs "embody law", that is, enforce a specific way to solve problems. The rules a computer follows are based on the assumptions of a computer programmer for how these problems might be solved. That means the code could incorporate the programmer's imagination of how the world works, including their biases and expectations. While a computer program can

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incorporate bias in this way, Weizenbaum also noted that any data fed to a machine additionally reflects "human decision making processes" as data is being selected. Finally, he noted that machines might also transfer good information with unintended consequences if users are unclear about how to interpret the results. Weizenbaum warned against trusting decisions made by computer programs that a user doesn't understand, comparing such faith to a tourist who can find his way to a hotel room exclusively by turning left or right on a coin toss. Crucially, the tourist has no basis of understanding how or why he arrived at his destination, and a successful arrival does not mean the process is accurate or reliable. An early example of algorithmic bias resulted in as many as 60 women and ethnic minorities denied entry to St. George's Hospital Medical School per year from 1982 to 1986, based on implementation of a new computer-guidance assessment system that denied entry to women and men with "foreign-sounding names" based on historical trends in admissions. While many schools at the time employed similar biases in their selection process, St. George was most notable for automating said bias through the use of an algorithm, thus gaining the attention of people on a much wider scale. In recent years, as algorithms increasingly rely on machine learning methods applied to real-world data, algorithmic bias has become more prevalent due to inherent biases within the data itself. For instance, facial recognition systems have been shown to misidentify individuals from marginalized groups at significantly higher rates than white individuals, highlighting how biases in training datasets manifest in deployed systems. A 2018 study by Joy Buolamwini and Timnit Gebru found that commercial facial recognition technologies exhibited error rates of up to 35% when identifying darker-skinned women, compared to less than 1% for lighter-skinned men. Algorithmic biases are not only technical failures but often reflect systemic inequities embedded in historical and societal data. Researchers and critics, such as Cathy O'Neil in her book *Weapons of Math Destruction* (2016), emphasize that these biases can amplify existing social inequalities under the guise of objectivity. O'Neil argues that opaque, automated decision-making processes in areas such as credit scoring, predictive policing, and education can reinforce discriminatory practices while appearing neutral or scientific.